

CS 314
Spring 2025
Final (100 points total)

Short Answer (5-7 sentences)

1. Describe the 4+1 model of software architecture. What is the importance of each of the views? Who would be the audience for such a model?
2. Pick one an architecture pattern that we did NOT discuss in class and describe it and when it should be used.
3. Discuss why a coding standard is important. What are some of the challenges with settling on one?
4. Compare and contrast validation testing and defect testing.

Multiple Choice

5. Which of the following is NOT a valid use of application architectures?
 - a. As a starting point for architectural design.
 - b. As a design checklist
 - c. As a way of bypassing low-level design work for a project
 - d. As a way of organizing the work of the development team
 - e. As a vocabulary for talking about application types

Essay (1-2 pages single spaced)

6. Discuss how the testing process is different in Agile than in plan-driven processes. I would put forth that the Agile testing practices help to produce better code. What could be done in plan-driven processes to help the testing process?
7. We talked a lot about reuse over the term. Describe the policies and processes you would put in place to govern reuse in the software development process. **Be sure to include reuse of software your company has written, commercial software, and open-source software.** Do your policies and processes change if you are in an Agile environment? Why or why not?